

Table D.1 CHF data of fluid R123 in straight horizontal tubes

Sr. No.	Tube Dia. mm	Heated Length mm	System Pressure bar	Inlet Temperature °C	Mass Flux kg/m ² s	Exit Quality	CHF kW/m ²
1	5.5	500	1.99	33.5	400.5	0.624	126.8
2	5.5	500	1.83	32.5	431.5	0.577	130.3
3	5.5	500	2.17	33.2	835.2	0.375	184.3
4	5.5	500	2.46	33.0	854.3	0.346	181.4
5	5.5	500	2.71	33.7	1054.9	0.299	210.1
6	5.5	500	2.48	33.4	1255.8	0.289	242.4
7	5.5	500	2.95	33.9	1281.7	0.260	243.4
8	5.5	500	2.65	34.1	1479.3	0.245	262.9
9	5.5	1000	2.02	32.8	416.4	0.810	83.8
10	5.5	1000	2.59	33.7	880.0	0.548	133.1
11	5.5	1000	2.59	33.7	898.0	0.501	126.7
12	5.5	1000	2.82	33.4	1055.9	0.485	149.6
13	5.5	1000	2.78	33.4	1078.7	0.443	142.4
14	5.5	1000	3.12	34.5	1312.1	0.427	173.4
15	5.5	1000	3.19	34.6	1327.1	0.450	183.0
16	7.5	500	1.83	31.7	448.1	0.546	158.4
17	7.5	500	2.05	32.3	665.9	0.462	209.4
18	7.5	500	2.04	29.3	682.7	0.389	217.9
19	7.5	500	2.08	28.2	687.5	0.282	172.2
20	7.5	500	2.17	30.4	979.0	0.328	247.8
21	7.5	500	2.39	31.0	1151.5	0.330	301.6
22	7.5	500	2.55	29.9	1162.0	0.251	290.3
23	7.5	500	2.37	31.0	1175.6	0.316	292.6
24	7.5	500	2.64	31.2	1619.8	0.160	328.3
25	7.5	500	2.92	31.2	1643.8	0.152	337.3
26	7.5	1000	2.04	29.9	381.8	0.849	107.0
27	7.5	1000	2.03	29.9	386.0	0.820	102.9
28	7.5	1000	2.32	29.5	531.8	0.722	130.2
29	7.5	1000	2.31	29.5	543.1	0.704	132.9
30	7.5	1000	2.20	29.0	684.1	0.492	143.8
31	7.5	1000	2.22	29.0	719.7	0.488	136.7
32	7.5	1000	2.19	29.0	720.3	0.461	130.7
33	7.5	1000	2.60	29.9	753.1	0.549	155.0
34	7.5	1000	2.54	29.8	763.9	0.492	144.0
35	7.5	1000	2.66	30.0	1152.9	0.337	177.0

Sr. No.	Tube Dia. mm	Heated Length mm	System Pressure bar	Inlet Temperature °C	Mass Flux Kg/m ² s	Exit Quality	CHF kW/m ²
36	7.5	1000	2.44	30.0	1163.2	0.350	180.5
37	9.5	500	1.62	28.8	406.2	0.197	89.4
38	9.5	500	1.65	29.6	427.6	0.187	88.4
39	9.5	500	1.77	29.2	712.3	0.158	139.3
40	9.5	500	1.85	29.4	714.5	0.202	170.9
41	9.5	500	2.35	32.1	1005.2	0.212	265.7
42	9.5	500	1.98	29.3	1007.5	0.247	286.0
43	9.5	500	2.36	29.1	1282.8	0.172	327.0
44	9.5	500	2.55	29.1	1289.2	0.162	328.9
45	9.5	1000	1.82	29.1	424.3	0.355	74.6
46	9.5	1000	1.71	30.3	432.5	0.364	74.5
47	9.5	1000	2.10	31.7	707.6	0.370	130.6
48	9.5	1000	2.70	31.4	1272.8	0.231	193.5
49	9.5	1000	2.69	31.5	1301.1	0.233	198.3
50	12	1000	1.67	29.5	262.6	0.518	78.1
51	12	1000	1.69	29.6	267.3	0.540	82.6
52	12	1000	1.82	31.2	350.4	0.438	90.8
53	12	1000	1.85	31.3	358.8	0.450	95.4
54	12	1000	1.96	33.1	435.2	0.371	98.5
55	12	1000	1.99	33.1	445.2	0.383	103.9

Table D.2 CHF data of water in helical coils

Sr. No.	Coil no.	Heated length	Mass Flux	System Pressure	Exit Quality	Heat Supply	Critical Heat Flux	CHF Ratio
		mm		kg/m ² s		bar		
1	Coil_1	503.5	568	4.33	0.410	22054	2324	1.260
2	Coil_1	503.5	569	4.38	0.434	22896	2412	1.352
3	Coil_1	503.5	576	4.11	0.374	21000	2213	1.175
4	Coil_1	503.5	589	3.97	0.363	21000	2213	1.172
5	Coil_1	503.5	590	4.25	0.387	22054	2324	1.227
6	Coil_1	503.5	753	4.91	0.356	27104	2856	1.465
7	Coil_1	503.5	760	4.83	0.311	25281	2664	1.324
8	Coil_1	503.5	779	4.83	0.332	26862	2830	1.405
9	Coil_1	503.5	811	4.91	0.276	25281	2664	1.161
10	Coil_1	503.5	855	4.75	0.242	24824	2616	1.005
11	Coil_1	503.5	929	4.45	0.207	24824	2616	0.894
12	Coil_1	722.7	272	2.64	0.809	16783	1232	1.979
13	Coil_1	722.7	338	3.22	0.762	20008	1469	2.082
14	Coil_1	722.7	352	3.04	0.746	20501	1505	2.089
15	Coil_1	722.7	455	3.94	0.675	24752	1817	1.989
16	Coil_1	722.7	555	4.75	0.661	30000	2202	2.387
17	Coil_1	722.7	592	4.70	0.604	30000	2202	1.962
18	Coil_1	722.7	743	5.60	0.528	34668	2545	1.693
19	Coil_1	722.7	744	5.48	0.519	34240	2513	1.626
20	Coil_1	722.7	784	5.42	0.480	34240	2513	1.455
21	Coil_1	722.7	878	5.57	0.389	33696	2474	1.155
22	Coil_1	722.7	881	5.48	0.391	33904	2489	1.191
23	Coil_1	722.7	1081	5.96	0.322	37496	2752	0.963
24	Coil_1	845	228	2.69	0.893	15300	961	2.020
25	Coil_1	845	235	2.62	0.862	15300	961	1.832
26	Coil_1	845	300	3.47	0.862	19686	1236	2.219
27	Coil_1	845	312	3.47	0.824	19686	1236	2.005
28	Coil_1	855	354	3.47	0.766	21141	1312	1.899
29	Coil_1	845	417	4.51	0.772	25249	1585	2.418
30	Coil_1	845	440	4.47	0.718	25249	1585	2.106
31	Coil_1	855	506	4.81	0.735	29646	1839	2.671
32	Coil_1	845	579	5.52	0.631	30550	1918	1.789
33	Coil_1	855	581	4.80	0.640	30738	1907	1.940

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Exit Quality	Critical		CHF Ratio
						Heat Supply W	Heat Flux kW/m ²	
34	Coil_1	845	593	6.31	0.743	35400	2223	3.322
35	Coil_1	845	629	6.23	0.670	34825	2186	2.519
36	Coil_1	845	635	6.16	0.661	34825	2186	2.464
37	Coil_1	845	640	6.32	0.668	35400	2223	2.580
38	Coil_1	845	657	6.36	0.644	35400	2223	2.407
39	Coil_1	855	696	5.95	0.615	36200	2246	2.463
40	Coil_1	855	736	6.30	0.638	39395	2444	3.133
41	Coil_1	1100	365	3.96	0.914	25150	1213	3.119
42	Coil_1	1100	365	3.96	0.949	25942	1251	5.504
43	Coil_1	1100	386	4.30	0.906	26476	1277	3.121
44	Coil_1	1100	386	4.30	0.937	27181	1311	4.743
45	Coil_1	1100	433	4.48	0.798	26873	1296	2.185
46	Coil_1	1100	433	4.48	0.903	29600	1428	3.661
47	Coil_1	1100	671	4.91	0.538	31416	1515	1.122
48	Coil_2	562	229	2.13	0.745	13140	1240	1.754
49	Coil_2	562	245	2.08	0.643	12478	1178	2.490
50	Coil_2	562	343	4.44	0.639	18020	1701	1.433
51	Coil_2	562	359	4.44	0.599	18020	1701	1.210
52	Coil_2	562	360	4.31	0.662	19381	1830	1.751
53	Coil_2	562	365	2.71	0.539	16482	1556	1.042
54	Coil_2	562	379	2.78	0.555	17514	1653	1.132
55	Coil_2	562	513	3.45	0.469	21344	2015	1.156
56	Coil_2	562	531	3.53	0.469	22134	2089	1.205
57	Coil_2	562	778	4.33	0.354	27560	2602	1.331
58	Coil_2	562	873	4.03	0.253	25400	2398	0.999
59	Coil_2	562	1165	4.74	0.210	31610	2984	0.895
60	Coil_2	562	1241	4.46	0.164	29960	2828	0.798
61	Coil_2	677	257	2.94	0.722	14559	1141	1.456
62	Coil_2	677	281	2.89	0.644	14559	1141	2.236
63	Coil_2	677	326	3.61	0.681	17784	1394	1.487
64	Coil_2	677	475	4.99	0.589	23718	1859	1.319
65	Coil_2	677	499	4.99	0.548	23718	1859	1.195
66	Coil_2	677	758	5.78	0.388	29174	2286	1.024
67	Coil_2	677	758	5.78	0.422	30669	2403	1.134
68	Coil_2	677	817	5.54	0.304	27232	2134	0.888

D. Experimental Critical Heat Flux Data

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Exit Quality	Heat Supply		Critical Heat Flux kW/m ²	CHF Ratio
						W			
69	Coil_2	764	311	2.77	0.773	18530		1287	1.855
70	Coil_2	764	316	2.71	0.685	17150		1191	1.333
71	Coil_2	764	493	3.90	0.647	26010		1806	1.714
72	Coil_2	764	528	3.86	0.568	25308		1757	1.254
73	Coil_2	764	768	4.77	0.444	31625		2196	1.287
74	Coil_2	764	791	4.60	0.395	30150		2094	1.120
75	Coil_2	764	987	5.36	0.373	36848		2559	1.172
76	Coil_2	764	1143	5.08	0.249	34010		2362	0.734
77	Coil_3	446.5	215	1.70	0.446	14535		1295	1.222
78	Coil_3	446.5	399	1.87	0.183	15584		1389	0.913
79	Coil_3	408.5	401	2.41	0.222	18054		1758	1.120
80	Coil_3	446.5	620	2.21	0.108	19740		1759	0.810
81	Coil_3	446.5	787	2.10	0.066	21153		1885	0.731
82	Coil_3	446.5	949	2.08	0.036	22261		1984	0.659
83	Coil_3	647.5	192	2.12	0.673	18040		1109	1.478
84	Coil_3	647.5	203	2.11	0.627	18040		1109	1.244
85	Coil_3	647.5	227	1.90	0.547	18011		1107	1.069
86	Coil_3	647.5	382	2.74	0.363	23375		1436	0.956
87	Coil_3	647.5	387	2.71	0.344	22848		1404	0.930
88	Coil_3	647.5	397	2.31	0.253	19125		1175	0.779
89	Coil_3	647.5	435	3.10	0.365	27000		1659	1.052
90	Coil_3	647.5	594	3.56	0.268	31164		1915	1.104
91	Coil_3	647.5	608	3.23	0.227	28820		1771	0.981
92	Coil_3	647.5	617	3.25	0.210	28171		1731	0.925
93	Coil_3	647.5	626	3.22	0.214	28820		1771	0.957
94	Coil_3	647.5	793	3.76	0.171	33660		2068	0.912
95	Coil_4	632	166	2.31	0.748	17072		1075	1.804
96	Coil_4	632	169	2.29	0.706	16556		1042	1.566
97	Coil_4	632	266	3.13	0.556	22050		1388	1.126
98	Coil_4	632	278	3.22	0.565	23391		1473	1.188
99	Coil_4	632	454	4.20	0.349	28244		1778	1.097
100	Coil_4	632	454	4.20	0.377	29625		1865	1.160
101	Coil_4	632	488	4.01	0.282	26760		1685	0.976

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Exit Quality	Critical	
						Heat Supply W	CHF Ratio
102	Coil_4	782	254	4.00	0.801	27987	2.522
103	Coil_4	782	303	3.99	0.610	27280	1.184
104	Coil_4	782	393	5.02	0.586	34765	1.408
105	Coil_4	782	436	4.93	0.432	31388	1.011
106	Coil_4	1500.5	265	3.60	0.922	32638	2.638
107	Coil_4	1500.5	270	3.51	0.833	30600	1.563
108	Coil_4	1500.5	272	3.51	0.824	30600	1.522
109	Coil_4	1500.5	478	4.36	0.444	34752	1.291
110	Coil_4	1500.5	488	4.37	0.431	34752	1.790
111	Coil_4	2294.5	293	3.27	0.569	24789	2.357
112	Coil_4	2294.5	294	3.27	0.566	24789	1.505
113	Coil_4	2294.5	300	3.27	0.551	24789	2.026
114	Coil_4	2294.5	505	3.23	0.243	24829	0.936
115	Coil_4	2294.5	505	3.24	0.243	24829	0.936
116	Coil_4	2294.5	507	3.23	0.241	24829	0.932
117	Coil_5	421.5	121	1.41	0.510	13178	1.393
118	Coil_5	421.5	144	1.48	0.435	13951	1.326
119	Coil_5	421.5	198	1.52	0.287	14416	1.129
120	Coil_5	421.5	276	1.48	0.158	14151	0.875
121	Coil_5	421.5	278	1.36	0.161	14151	0.899
122	Coil_5	421.5	321	1.47	0.118	14310	0.717
123	Coil_5	421.5	326	1.22	0.124	14310	0.769
124	Coil_5	421.5	374	1.49	0.100	15615	0.710
125	Coil_5	421.5	378	1.51	0.097	15615	0.701
126	Coil_5	421.5	428	1.50	0.076	16207	0.675
127	Coil_5	421.5	490	1.52	0.064	17641	0.692
128	Coil_5	421.5	529	1.42	0.051	17612	0.671
129	Coil_5	421.5	623	1.49	0.031	18952	0.638
130	Coil_5	421.5	669	1.38	0.028	19542	0.652
131	Coil_5	421.5	830	1.25	0.001	19740	0.578
132	Coil_5	617.5	273	2.33	0.338	23089	1.000
133	Coil_5	617.5	421	2.46	0.167	24167	0.815
134	Coil_5	617.5	428	2.51	0.174	25129	0.849

Sr. No.	Coil no.	Heated length	Mass Flux	System Pressure	Exit Quality	Heat Supply	Critical	
							W	Heat Flux
		mm	kg/m ² s	bar				kW/m ²
135	Coil_5	1059	121	1.72	0.862	20264	628	1.672
136	Coil_5	1059	158	2.20	0.806	25217	781	1.732
137	Coil_5	1059	203	2.68	0.748	30775	954	1.702
138	Coil_5	2084	92	1.73	0.949	18407	290	2.526
139	Coil_5	2084	92	1.71	0.955	18170	286	2.456
140	Coil_5	658	128	2.09	0.755	19450	970	1.888
141	Coil_5	658	266	3.17	0.417	26618	1327	1.040
142	Coil_5	658	275	2.69	0.390	25884	1283	1.010
143	Coil_5	658	470	3.95	0.228	33790	1685	0.984
144	Coil_5	1406	140	2.24	0.921	24960	1221	2.102
145	Coil_5	1406	144	2.30	0.924	25845	603	2.271
146	Coil_5	1406	145	2.27	0.882	24960	1224	1.579
147	Coil_5	1406	145	2.26	0.878	24960	1222	1.560
148	Coil_6	678	192	2.50	0.526	23575	1107	1.140
149	Coil_6	585	281	2.80	0.322	24960	1358	1.032
150	Coil_6	678	320	3.34	0.359	30940	1453	1.060
151	Coil_6	765	410	3.12	0.246	31608	1315	0.877
152	Coil_6	1152	189	2.74	0.753	30615	846	1.562
153	Coil_6	1152	290	3.45	0.522	36120	998	1.897
154	Coil_6	1749	188	2.92	0.811	32364	589	2.790
155	Coil_6	1749	311	3.38	0.416	33168	604	1.241
156	Coil_6	1749	442	3.41	0.234	33562	611	0.920

Table D.3 CHF data of fluid R123 at low pressure in helical coils

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
1	Coil_1	2146	284	2.11	108.8	1.084	3844	60.0
2	Coil_1	2146	349	2.36	97.2	1.076	4737	74.0
3	Coil_1	2146	422	2.58	88.6	1.073	5765	90.0
4	Coil_1	2146	496	2.82	80.5	1.026	6585	102.8
5	Coil_1	2146	570	3.07	73.7	1.001	7482	116.8
6	Coil_1	1524	436	2.28	100.8	0.588	3538	77.8
7	Coil_1	1524	568	2.59	88.3	0.580	4666	102.6
8	Coil_1	1524	706	2.85	79.6	0.574	5902	129.8
9	Coil_1	1524	867	3.27	68.9	0.515	6905	151.8
10	Coil_1	1524	1012	3.63	61.6	0.498	8078	177.6
11	Coil_1	1006	145	1.74	131.7	1.087	1947	64.9
12	Coil_1	1006	287	1.93	119.0	0.649	2436	81.1
13	Coil_1	1006	425	2.12	108.3	0.523	3060	101.9
14	Coil_1	1006	558	2.27	100.9	0.452	3641	121.3
15	Coil_1	1006	701	2.44	93.8	0.388	4159	138.5
16	Coil_1	1006	710	2.61	87.5	0.303	3641	121.3
17	Coil_1	1006	851	2.59	88.2	0.318	4483	149.3
18	Coil_1	1006	987	2.71	84.0	0.284	4899	163.2
19	Coil_1	504	423	1.85	123.9	0.163	1213	80.7
20	Coil_1	504	560	1.96	117.3	0.136	1483	98.6
21	Coil_1	504	699	2.10	109.6	0.194	2367	157.4
22	Coil_1	504	704	2.06	111.6	0.122	1807	120.1
23	Coil_1	504	851	2.09	109.8	0.098	2001	133.0
24	Coil_1	504	984	2.18	105.3	0.118	2570	170.9
25	Coil_1	504	994	2.13	107.7	0.075	2140	142.2
26	Coil_1	504	1127	2.27	101.3	0.067	2436	162.0
27	Coil_1	504	1276	2.39	95.9	0.055	2682	178.3
28	Coil_1	610	427	1.91	120.4	0.193	1406	77.2
29	Coil_1	610	704	2.14	107.1	0.146	2116	116.2
30	Coil_1	610	983	2.19	104.6	0.137	3030	166.4
31	Coil_1	610	992	2.38	96.4	0.103	2754	151.3

D. Experimental Critical Heat Flux Data

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
32	Coil_2	1420	432	2.35	97.7	1.003	3513	105.0
33	Coil_2	1420	435	2.36	97.0	1.039	3638	108.7
34	Coil_2	1420	579	2.56	89.3	0.983	4653	139.1
35	Coil_2	1420	707	3.21	70.1	0.886	5359	160.2
36	Coil_2	937	318	1.94	118.8	0.856	2175	98.5
37	Coil_2	937	328	1.97	116.8	0.870	2278	103.2
38	Coil_2	937	370	2.14	107.3	0.897	2700	122.3
39	Coil_2	937	372	2.14	107.5	1.056	3125	141.5
40	Coil_2	937	454	2.23	102.8	0.893	3296	149.3
41	Coil_2	937	510	2.44	93.8	0.786	3357	152.0
42	Coil_2	937	558	2.76	82.3	0.721	3485	157.9
43	Coil_2	937	674	2.69	84.7	0.689	4101	185.8
44	Coil_2	937	775	2.74	83.2	0.619	4334	196.3
45	Coil_2	937	883	2.87	78.9	0.565	4641	210.2
46	Coil_2	937	974	2.98	75.9	0.544	4998	226.4
47	Coil_2	937	1070	3.27	68.7	0.495	5233	237.0
48	Coil_2	467	206	1.80	127.7	0.718	1195	108.6
49	Coil_2	467	337	1.92	119.8	0.455	1333	121.1
50	Coil_2	467	448	1.74	131.5	0.345	1386	126.0
51	Coil_2	467	457	2.12	108.6	0.331	1435	130.4
52	Coil_2	467	579	2.12	108.2	0.287	1673	152.1
53	Coil_2	467	684	2.25	102.1	0.231	1794	163.0
54	Coil_2	467	825	2.38	96.3	0.196	2003	182.1
55	Coil_2	467	909	2.40	95.3	0.180	2092	190.1
56	Coil_2	467	1130	2.62	87.1	0.138	2350	213.5
57	Coil_2	467	1136	2.15	106.7	0.131	2158	196.1
58	Coil_2	467	1620	2.37	96.7	0.057	2352	213.8
59	Coil_3	1472	421	2.36	97.3	1.169	2103	82.7
60	Coil_3	1472	527	2.62	87.0	1.034	2387	93.8
61	Coil_3	1472	643	2.91	77.9	1.004	2876	113.1
62	Coil_3	1472	694	3.04	74.5	0.972	3042	119.6
63	Coil_3	1472	774	3.28	68.6	0.912	3255	128.0
64	Coil_3	482	435	2.04	112.6	0.615	1172	140.7
65	Coil_3	482	658	2.25	101.8	0.507	1571	188.7

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
66	Coil_3	482	757	2.37	96.7	0.483	1778	213.5
67	Coil_3	482	845	2.48	92.1	0.458	1950	234.1
68	Coil_3	482	1060	2.81	81.0	0.420	2311	277.5
69	Coil_3	482	1269	3.05	74.0	0.374	2627	315.4
70	Coil_3	482	1466	3.27	68.8	0.328	2861	343.5
71	Coil_3	970	254	1.93	118.8	1.129	1203	71.8
72	Coil_3	970	468	2.46	92.9	0.826	1735	103.5
73	Coil_3	970	647	3.06	73.9	0.772	2355	140.5
74	Coil_3	970	746	3.34	67.2	0.726	2633	157.1
75	Coil_3	970	854	3.68	60.8	0.695	2975	177.5
76	Coil_4	1508	421	2.35	97.3	0.916	4950	110.0
77	Coil_4	1508	498	2.57	88.7	0.813	5333	118.5
78	Coil_4	1508	574	2.72	83.6	0.811	6171	137.1
79	Coil_4	1508	725	3.10	72.8	0.707	7106	157.9
80	Coil_4	989	428	1.85	124.4	0.406	2360	80.0
81	Coil_4	989	564	2.06	111.3	0.357	2899	98.2
82	Coil_4	989	707	2.24	102.6	0.297	3262	110.5
83	Coil_4	989	856	2.41	94.8	0.281	3895	132.0
84	Coil_4	989	1000	2.63	86.7	0.251	4367	148.0
85	Coil_4	989	1002	2.66	85.5	0.255	4451	150.8
86	Coil_4	989	1141	2.71	84.1	0.224	4708	159.5
87	Coil_4	989	1283	2.92	77.7	0.198	5152	174.5
88	Coil_4	506	425	1.84	124.6	0.237	1524	100.9
89	Coil_4	506	578	1.90	121.0	0.188	1775	117.5
90	Coil_4	506	704	2.02	113.7	0.142	1836	121.5
91	Coil_4	506	715	2.00	115.0	0.145	1919	127.1
92	Coil_4	506	853	2.07	110.8	0.130	2182	144.5
93	Coil_4	506	988	2.25	101.8	0.108	2433	161.1
94	Coil_4	506	990	2.24	102.6	0.104	2380	157.6
95	Coil_4	506	1132	2.33	98.3	0.097	2740	181.4
96	Coil_4	506	1273	2.50	91.5	0.080	3036	201.0
97	Coil_4	506	1274	2.45	93.3	0.075	2928	193.9
98	Coil_5	1529	457	2.40	95.2	0.993	3616	100.4

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
99	Coil_5	1529	467	2.41	94.8	0.991	3696	102.6
100	Coil_5	1529	568	2.68	84.9	0.929	4296	119.2
101	Coil_5	1529	690	2.99	75.6	0.909	5198	144.3
102	Coil_5	1006	233	1.84	124.6	1.031	1867	78.8
103	Coil_5	1006	241	1.87	122.7	1.026	1928	81.3
104	Coil_5	1006	347	2.10	109.1	0.964	2639	111.3
105	Coil_5	1006	453	2.34	98.0	0.886	3216	135.7
106	Coil_5	1006	573	2.49	91.8	0.753	3583	151.2
107	Coil_5	1006	693	2.65	86.2	0.684	4068	171.6
108	Coil_5	1006	819	2.86	79.2	0.632	4579	193.2
109	Coil_5	1006	941	3.16	71.5	0.643	5423	228.8
110	Coil_5	1006	1078	3.19	70.7	0.585	5897	248.8
111	Coil_5	1006	1195	3.48	64.4	0.528	6317	266.5
112	Coil_5	513	233	1.80	127.4	0.472	919	76.0
113	Coil_5	513	241	1.80	127.8	0.469	946	78.3
114	Coil_5	513	472	2.10	109.4	0.328	1461	120.9
115	Coil_5	513	584	2.21	103.7	0.280	1642	135.9
116	Coil_5	513	696	2.33	98.5	0.250	1851	153.1
117	Coil_5	513	712	2.35	97.6	0.244	1872	154.8
118	Coil_5	513	820	2.44	93.7	0.227	2102	173.9
119	Coil_5	513	929	2.63	86.6	0.214	2393	198.0
120	Coil_5	513	1027	2.79	81.4	0.197	2600	215.1
121	Coil_5	513	1152	2.98	75.9	0.196	3008	248.8
122	Coil_5	513	1385	3.13	72.1	0.149	3272	270.7
123	Coil_5	513	1620	3.30	68.1	0.114	3573	295.6
124	Coil_6	1472	449	2.27	101.0	0.808	1540	61.7
125	Coil_6	1472	469	2.33	98.2	0.848	1685	67.5
126	Coil_6	1472	692	2.64	86.2	0.711	2194	87.8
127	Coil_6	1472	824	3.05	74.1	0.639	2484	99.5
128	Coil_6	1472	946	3.38	66.5	0.572	2688	107.6
129	Coil_6	984	427	2.04	112.8	0.640	1191	71.4
130	Coil_6	984	546	2.23	103.0	0.583	1438	86.1
131	Coil_6	984	675	2.48	92.2	0.561	1775	106.3
132	Coil_6	984	884	2.76	82.6	0.464	2037	122.0

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
133	Coil_6	984	1092	2.82	80.7	0.426	2426	145.3
134	Coil_6	984	1263	3.28	68.6	0.375	2731	163.6
135	Coil_6	984	1406	3.62	61.7	0.330	2933	175.7
136	Coil_6	476	443	2.19	104.8	0.502	1038	128.6
137	Coil_6	476	661	2.60	87.8	0.363	1277	158.2
138	Coil_6	476	888	2.92	77.7	0.256	1457	180.4
139	Coil_6	476	1182	3.07	73.7	0.200	1765	218.6
140	Coil_6	476	1444	3.27	68.8	0.145	1990	246.4

Table D.4 CHF data of fluid R123 at high pressure in helical coils

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
1	Coil_1	2146	289	5.36	40.6	1.077	4456	69.6
2	Coil_1	2146	362	5.42	40.0	1.072	5545	86.6
3	Coil_1	2146	437	5.51	39.4	1.031	6506	101.6
4	Coil_1	2146	493	2.76	82.4	1.004	6624	103.4
5	Coil_1	2146	502	5.64	38.4	0.998	7308	114.1
6	Coil_1	2146	581	5.68	38.1	0.990	8408	131.3
7	Coil_1	2146	718	5.73	37.7	0.897	9688	151.3
8	Coil_1	1524	434	5.61	38.6	0.841	5603	123.2
9	Coil_1	1524	579	5.86	36.8	0.866	7650	168.2
10	Coil_1	1524	726	6.13	35.0	0.854	9551	210.0
11	Coil_1	1524	890	6.44	33.3	0.724	10650	234.1
12	Coil_1	1524	975	6.59	32.4	0.678	11273	247.8
13	Coil_1	1524	1069	6.80	31.3	0.654	12186	267.9
14	Coil_1	1006	147	5.06	43.1	0.356	1133	37.7
15	Coil_1	1006	263	5.13	42.5	0.540	2556	85.1
16	Coil_1	1006	436	5.39	40.3	0.401	3711	123.6
17	Coil_1	1006	440	5.24	41.6	0.428	3776	125.8
18	Coil_1	1006	578	5.65	38.3	0.408	4975	165.7
19	Coil_1	1006	719	5.93	36.4	0.475	6699	223.1
20	Coil_1	1006	823	5.87	36.8	0.407	7140	237.8

D. Experimental Critical Heat Flux Data

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
21	Coil_1	1006	948	5.96	36.2	0.341	7626	254.0
22	Coil_1	1006	1073	6.06	35.5	0.323	8460	281.8
23	Coil_1	610	186	5.22	41.7	0.070	932	51.2
24	Coil_1	610	277	5.24	41.6	0.099	1475	81.0
25	Coil_1	610	439	5.42	40.1	0.392	3701	203.3
26	Coil_1	610	578	5.44	39.9	0.230	3905	214.5
27	Coil_1	610	718	5.47	39.7	0.179	4451	244.5
28	Coil_1	610	854	5.49	39.5	0.107	4635	254.6
29	Coil_1	610	992	5.53	39.2	0.077	5039	276.8
30	Coil_2	1420	483	4.55	48.4	1.020	4235	126.6
31	Coil_2	1420	613	4.65	47.2	1.013	5394	161.2
32	Coil_2	1420	744	4.75	46.2	0.962	6347	189.7
33	Coil_2	1420	904	5.20	41.9	0.729	6398	191.2
34	Coil_2	1420	943	5.37	40.5	0.689	6466	193.2
35	Coil_2	1420	1057	5.46	39.7	0.641	6994	209.0
36	Coil_2	1420	1184	5.56	39.0	0.566	7353	219.8
37	Coil_2	937	343	4.82	45.5	1.008	3042	137.8
38	Coil_2	937	448	4.66	47.1	0.981	3904	176.8
39	Coil_2	937	591	4.86	45.0	0.944	5039	228.2
40	Coil_2	937	675	2.71	84.2	0.434	3037	137.6
41	Coil_2	937	733	5.06	43.1	0.851	5859	265.4
42	Coil_2	937	956	5.45	39.8	0.689	6750	305.7
43	Coil_2	937	1198	5.97	36.1	0.510	7276	329.6
44	Coil_2	937	1631	5.79	37.3	0.278	7365	333.6
45	Coil_2	467	207	4.92	44.5	0.372	998	90.7
46	Coil_2	467	344	5.05	43.2	0.457	1887	171.5
47	Coil_2	467	479	5.18	42.0	0.360	2378	216.1
48	Coil_2	467	694	5.27	41.3	0.226	2884	262.1
49	Coil_2	467	921	5.22	41.7	0.097	2996	272.3
50	Coil_2	467	1151	5.19	42.0	0.047	3290	299.0
51	Coil_2	467	1356	5.17	42.1	-0.006	3432	311.9
52	Coil_2	467	1568	5.17	42.2	-0.056	3482	316.5
53	Coil_2	467	1569	5.16	42.3	-0.048	3562	323.8

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
54	Coil_3	1472	378	4.86	45.0	1.097	1939	76.2
55	Coil_3	1472	624	5.61	38.6	1.023	3102	121.9
56	Coil_3	1472	809	5.69	38.0	1.016	3987	156.8
57	Coil_3	1472	1021	6.16	34.9	0.917	4761	187.2
58	Coil_3	1472	1060	3.80	58.6	0.665	3650	143.5
59	Coil_3	1472	1113	6.21	34.6	0.969	5377	211.4
60	Coil_3	1472	1422	6.75	31.6	0.774	6041	237.5
61	Coil_3	970	628	5.54	39.2	0.979	3021	180.2
62	Coil_3	970	820	6.04	35.6	0.897	3769	224.9
63	Coil_3	970	823	6.05	35.5	0.927	3867	230.7
64	Coil_3	970	1136	6.56	32.6	0.718	4607	274.9
65	Coil_3	970	1281	6.56	32.6	0.651	4906	292.7
66	Coil_3	970	1463	6.67	32.0	0.550	5130	306.1
67	Coil_3	482	402	5.02	43.5	0.856	1743	209.2
68	Coil_3	482	636	5.26	41.4	0.660	2344	281.4
69	Coil_3	482	827	5.50	39.4	0.530	2705	324.8
70	Coil_3	482	1091	5.23	41.6	0.365	2890	346.9
71	Coil_3	482	1486	5.48	39.6	0.200	3173	380.9
72	Coil_3	482	1773	5.48	39.6	0.119	3300	396.2
73	Coil_3	482	2062	5.47	39.7	0.070	3497	419.8
74	Coil_5	1529	446	5.03	43.4	1.035	4106	114.0
75	Coil_5	1529	574	5.25	41.5	0.916	4876	135.3
76	Coil_5	1529	704	5.47	39.7	0.943	6130	170.1
77	Coil_5	1529	777	3.18	71.0	0.904	6143	170.5
78	Coil_5	1529	922	5.76	37.6	0.856	7469	207.3
79	Coil_5	1529	1040	5.85	36.9	0.740	7694	213.6
80	Coil_5	1529	1199	5.87	36.7	0.604	7850	217.9
81	Coil_5	1006	461	4.84	45.2	0.959	3998	168.7
82	Coil_5	1006	592	5.05	43.2	0.824	4636	195.6
83	Coil_5	1006	709	5.24	41.5	0.798	5459	230.3
84	Coil_5	1006	941	5.35	40.6	0.576	5977	252.2
85	Coil_5	1006	1182	5.43	40.0	0.486	6891	290.7
86	Coil_5	1006	1378	5.48	39.6	0.379	7067	298.1
87	Coil_5	1006	1610	5.48	39.6	0.305	7428	313.4

D. Experimental Critical Heat Flux Data

Sr. No.	Coil no.	Heated length mm	Mass Flux kg/m ² s	System Pressure bar	Density Ratio	Exit Quality	Heat Supply	
							Watt	CHF kW/m ²
88	Coil_5	513	460	4.85	45.2	0.386	2287	189.2
89	Coil_5	513	657	4.99	43.8	0.320	3040	251.5
90	Coil_5	513	659	4.98	43.9	0.299	2955	244.4
91	Coil_5	513	914	5.14	42.4	0.225	3713	307.2
92	Coil_5	513	1138	5.27	41.3	0.162	4203	347.7
93	Coil_5	513	1374	5.16	42.3	0.081	4301	355.8
94	Coil_5	513	1638	5.02	43.5	0.018	4361	360.8
95	Coil_6	1472	409	5.06	43.2	1.096	2023	81.0
96	Coil_6	1472	681	5.38	40.4	0.997	3183	127.5
97	Coil_6	1472	928	5.84	37.0	0.944	4204	168.4
98	Coil_6	1472	940	5.80	37.3	0.929	4194	168.0
99	Coil_6	1472	1041	6.00	35.9	0.879	4512	180.7
100	Coil_6	1472	1155	6.17	34.8	0.807	4752	190.3
101	Coil_6	984	391	4.98	43.9	1.025	1841	110.3
102	Coil_6	984	625	5.06	43.1	0.951	2811	168.4
103	Coil_6	984	947	5.63	38.4	0.799	3809	228.2
104	Coil_6	984	1120	6.00	35.9	0.743	4367	261.6
105	Coil_6	984	1175	5.04	43.3	0.817	4734	283.6
106	Coil_6	984	1376	6.03	35.7	0.671	5066	303.4
107	Coil_6	984	1477	6.13	35.0	0.621	5209	312.0
108	Coil_6	984	1719	6.37	33.7	0.565	5800	347.4
109	Coil_6	984	2029	6.72	31.7	0.445	6159	369.0
110	Coil_6	476	360	4.88	44.9	0.769	1400	173.4
111	Coil_6	476	611	5.18	42.1	0.440	1739	215.3
112	Coil_6	476	804	5.38	40.4	0.341	2054	254.4
113	Coil_6	476	1058	5.66	38.3	0.248	2390	295.9
114	Coil_6	476	1062	5.94	36.3	0.230	2390	295.9
115	Coil_6	476	1277	5.43	40.0	0.154	2465	305.3
116	Coil_6	476	1531	5.63	38.4	0.126	2848	352.7
117	Coil_6	476	1765	5.83	37.1	0.095	3143	389.2